

Data Science 2025

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Research Question

To what extent do the Spotify audio features 'danceability' and 'energy' predict a song's commercial success?

AGENDA

**Spotify in a
Business context**

**Data Sources +
Challenges**

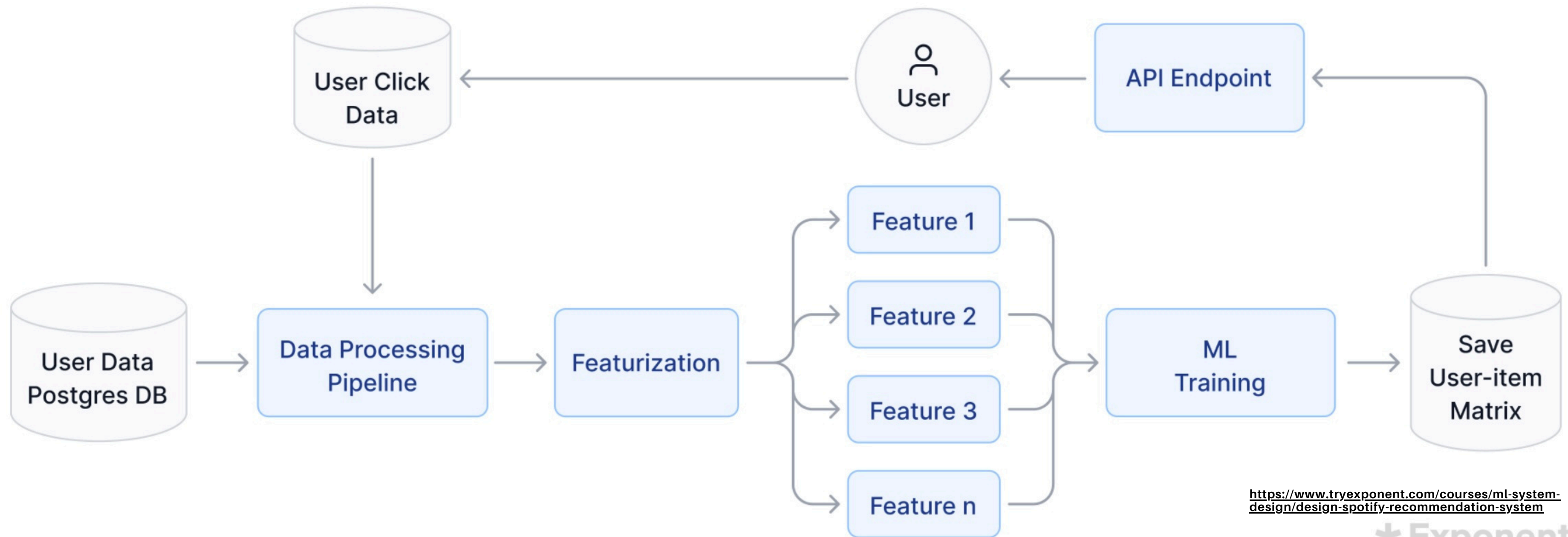
**Exploratory Data
Analysis**

First conclusion

Limitations

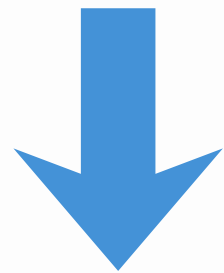
**Next Steps
(final Report)**

Spotify relevance in a Business context



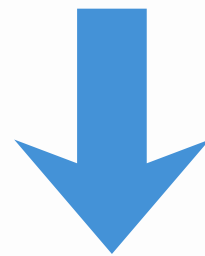
<https://www.tryexponent.com/courses/ml-system-design/design-spotify-recommendation-system>

Data sources + used variables



- ✗ Problem: A track ID is required for every single song → extremely time-consuming
- ✗ Tokens expire every 60 minutes → API repeatedly stops working

Conclusion: API unsuitable for large datasets



- ✗ Python Code
- ✓ Originally available only in Python → had to be converted to R & Quarto for our project



- ✓ Used only for general overview, not suitable for analysis



- ✓ Useful additional data (genre, release year)
- ✗ Inconsistent data quality

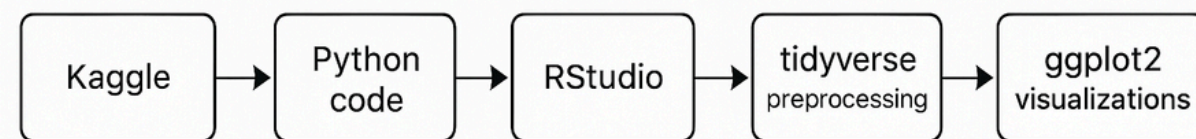
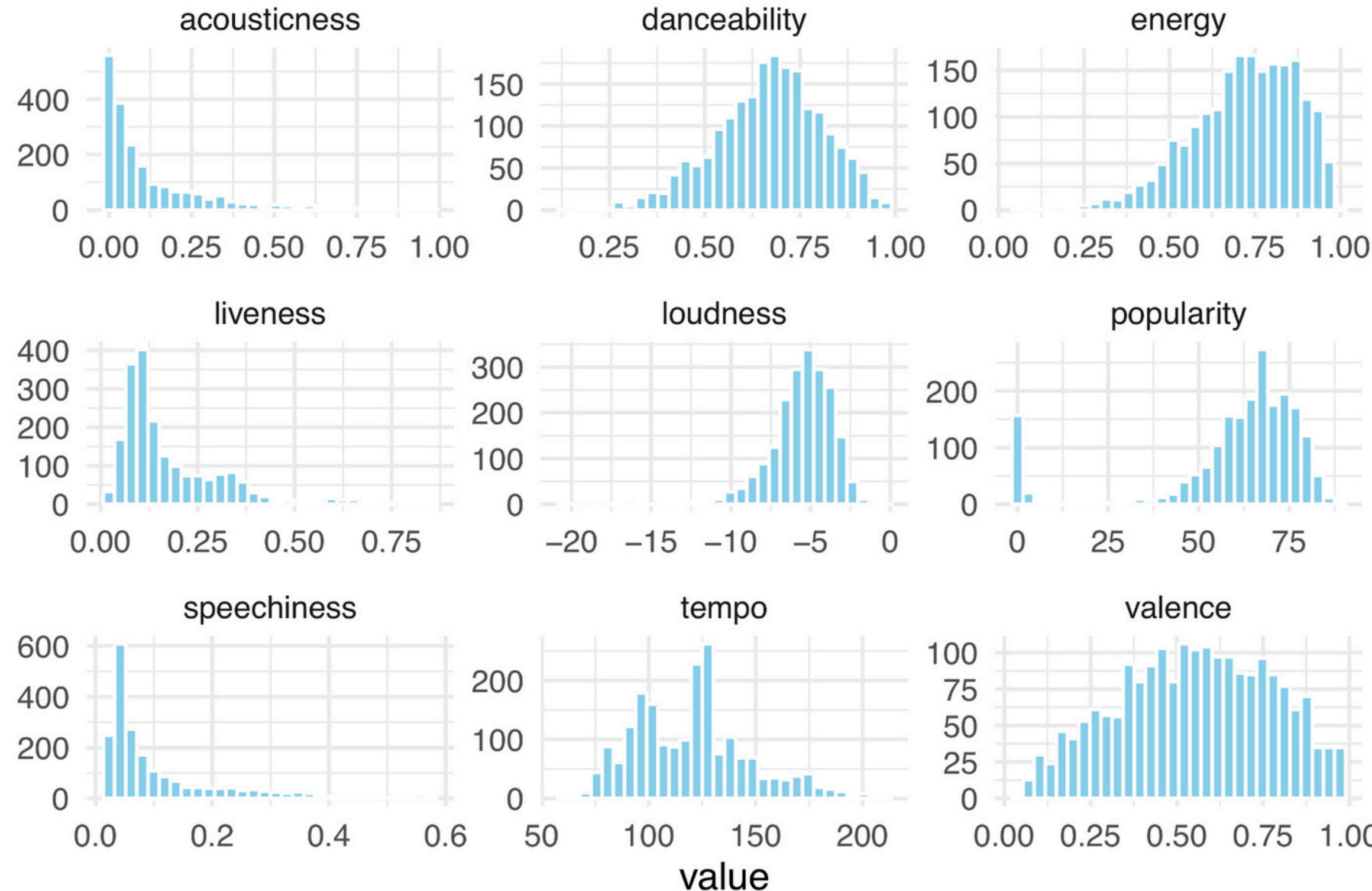


- ✓ Good overview of real chart data
- ✗ no audio features → not usable for analysis

Distribution of Spotify Audio Features

- Many features are skewed → strong clustering
- Energy & danceability show wide variation
- Popularity is widely spread → good for prediction analysis

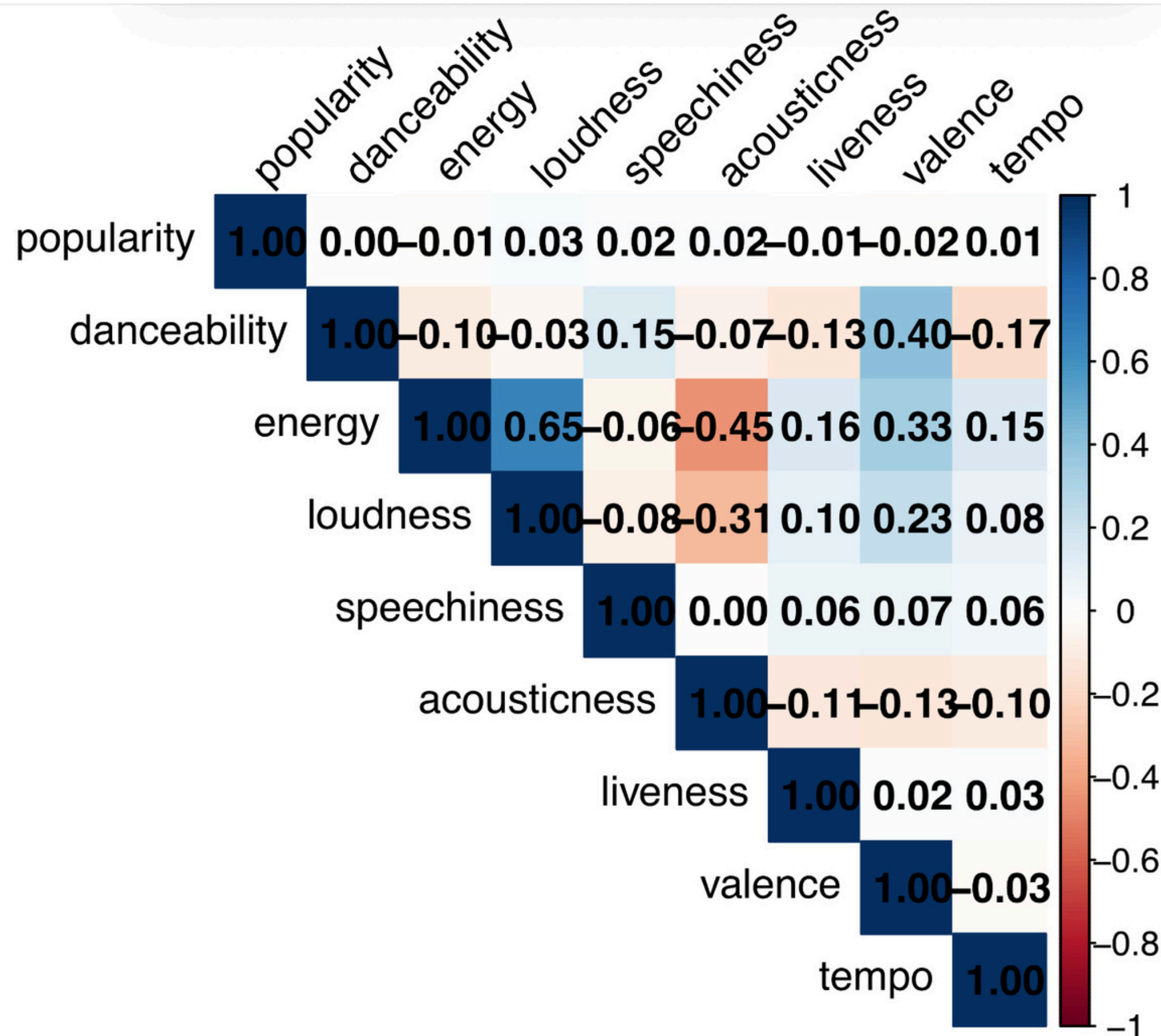
Feature Distribution



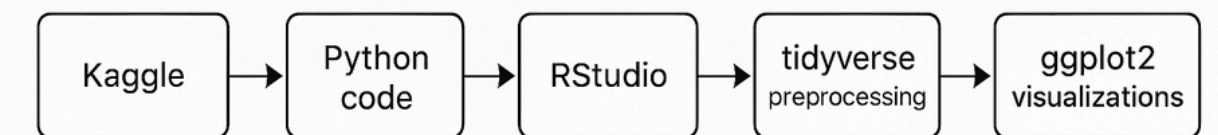
```

+ > features_long <- spotify %>%
+   select(popularity, danceability, energy, loudness, speechiness,
+     acousticness, liveness, valence, tempo) %>%
+     + pivot_longer(cols = everything(),
+       +       names_to = "feature",
+       +       values_to = "value")
+ > > ggplot(features_long, aes(x = value)) +
+   geom_histogram(bins = 30, fill = "skyblue", color = "white") +
+   + facet_wrap(~ feature, scales = "free") +
+   +   theme_minimal() +
+   +   labs(title = "Feature Distribution")
  
```


Correlation between Spotify Audio Features and Popularity



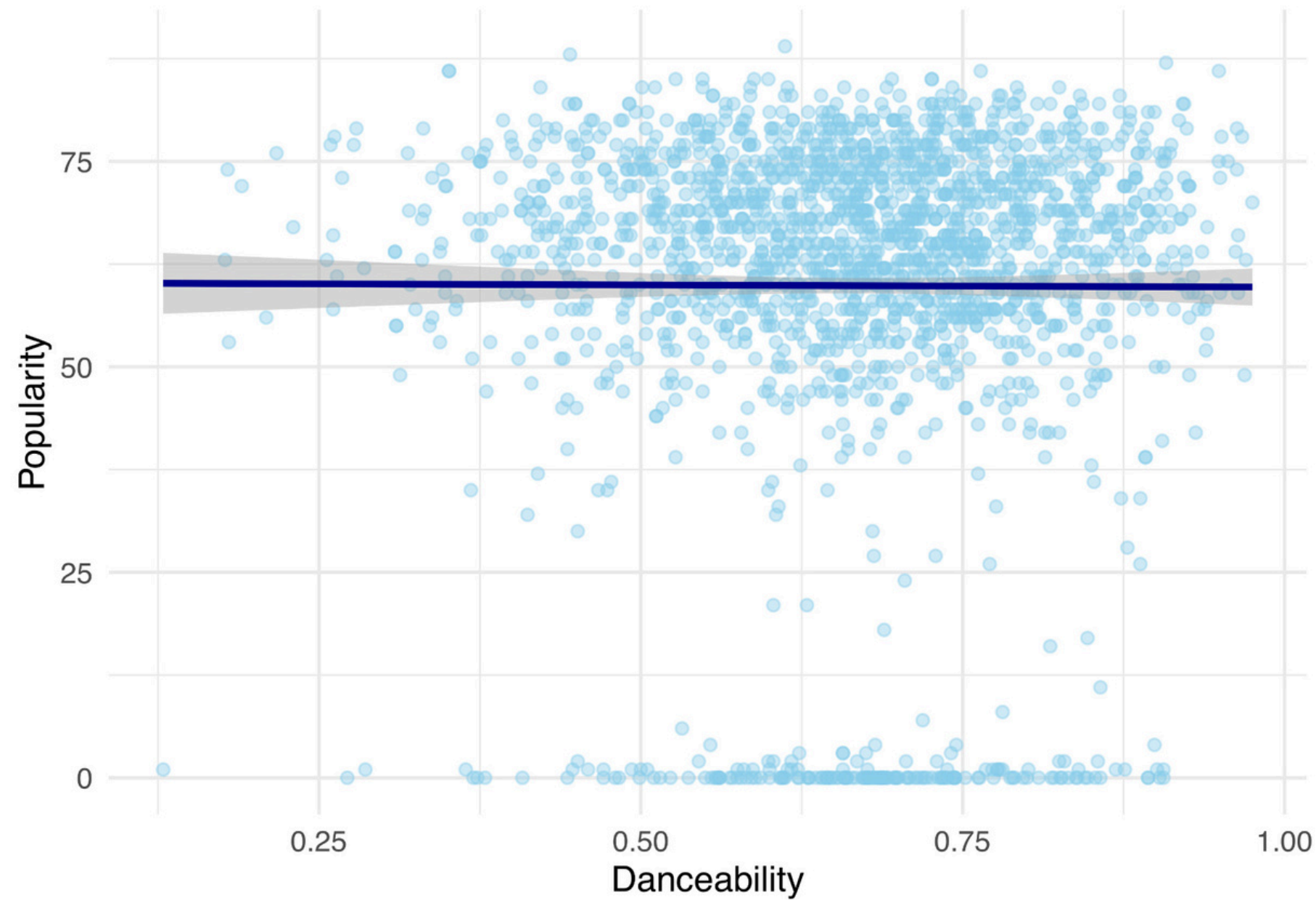
- Energy shows the strongest positive correlation with popularity
- Danceability also has a slight positive correlation with popularity
- Acousticness is negatively correlated with both energy and popularity
- Valence (positiveness) shows almost no correlation with popularity → a "happy vibe" alone does not predict a hit



```
spotify_numeric <- spotify %>%  
+ select(popularity, danceability, energy, loudness, speechiness,  
+        acousticness, liveness, valence, tempo)  
> corr_matrix <- cor(spotify_numeric, use = "complete.obs")  
+ > corrplot(corr_matrix,  
+           method = "color",  
+           type = "upper",  
+           addCoef.col = "black",  
+           tl.col = "black",  
+           tl.srt = 45)
```

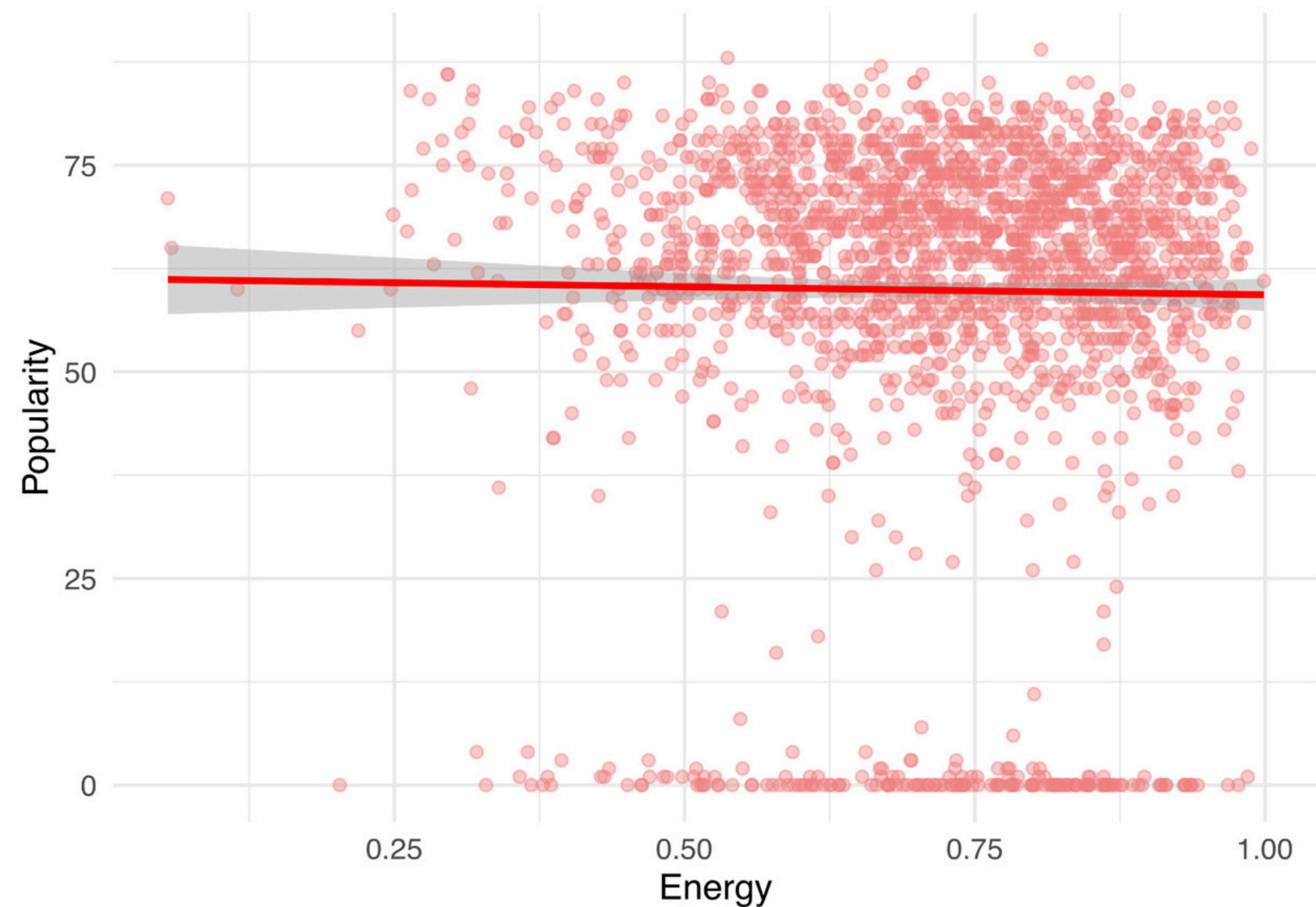

Scatterplot Insights

Relationship Between Danceability and Popularity



```
ggplot(spotify, aes(x = danceability, y = popularity)) +  
  geom_point(alpha = 0.4, color = "skyblue") +  
  geom_smooth(method = "lm", color = "darkblue") +  
  theme_minimal() +  
  labs(  
    title = "Relationship Between Danceability and Popularity",  
    x = "Danceability",  
    y = "Popularity"  
  )
```

Relationship Between Energy and Popularity

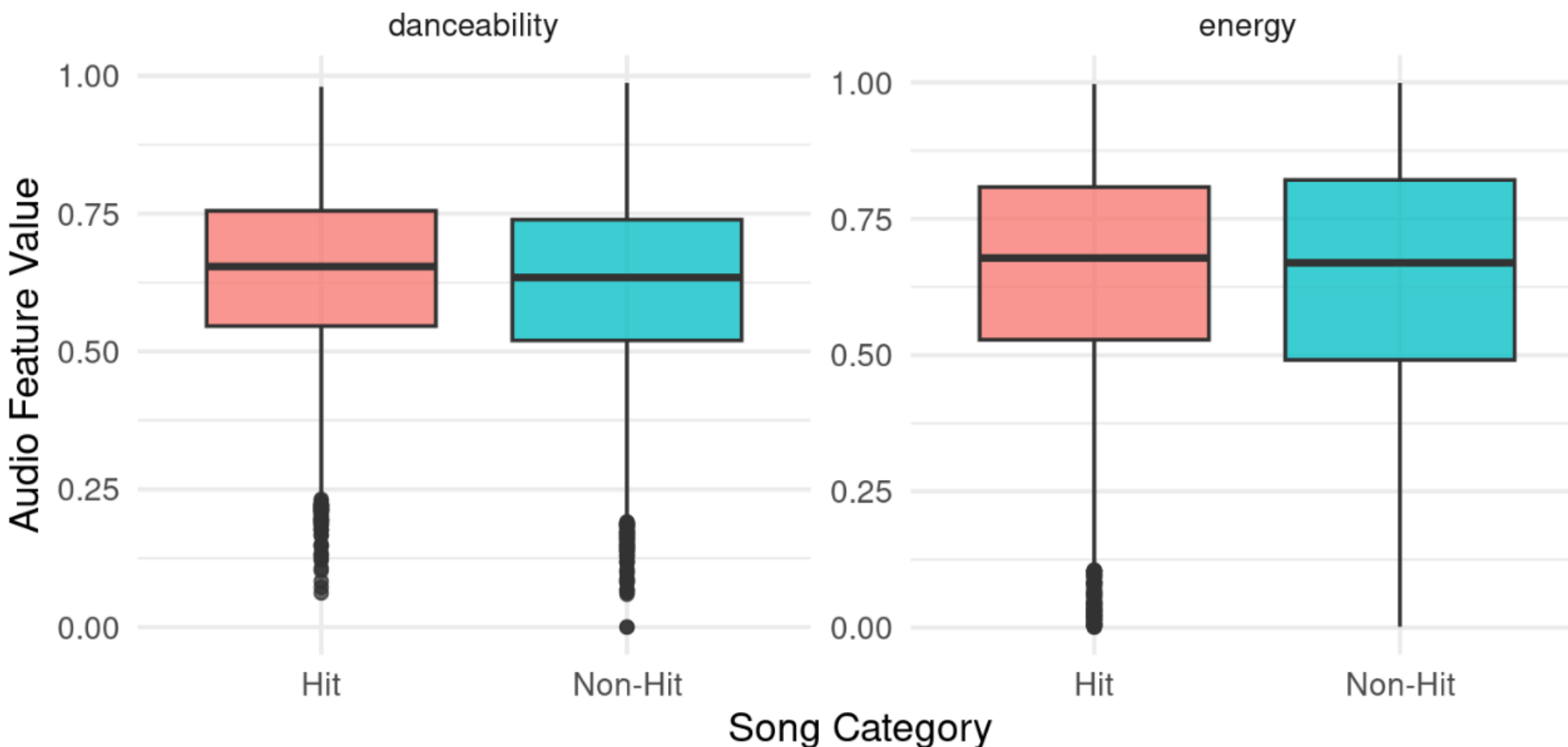


```
ggplot(spotify, aes(x = energy, y = popularity)) +  
  geom_point(alpha = 0.4, color = "lightcoral") +  
  geom_smooth(method = "lm", color = "red") +  
  theme_minimal() +  
  labs(  
    title = "Relationship Between Energy and Popularity",  
    x = "Energy",  
    y = "Popularity"  
  )
```

Boxplot comparison

Boxplot Comparison: What Makes a Hit?

How Danceability & Energy differ between Hit and Non-Hit Songs



```
library(tidyverse)

# 1. derive Hit vs. Non-Hit from Popularity (Median-Split)
threshold <- median(song_data$song_popularity, na.rm = TRUE)

song_data <- song_data %>%
  mutate(
    hit = if_else(song_popularity >= threshold, "Hit", "Non-Hit")
  )

# 2. Data in Long-Format for two Boxplots (Danceability & Energy)
song_data_long <- song_data %>%
  pivot_longer(
    cols = c(danceability, energy),
    names_to = "feature",
    values_to = "value"
  )

# 3. draw Boxplot
ggplot(song_data_long, aes(x = hit, y = value, fill = hit)) +
  geom_boxplot(alpha = 0.75) +
  facet_wrap(~ feature, scales = "free_y") +
  labs(
    title = "Boxplot Comparison: What Makes a Hit?",
    subtitle = "How Danceability & Energy differ between Hit and Non-Hit Songs",
    x = "Song Category",
    y = "Audio Feature Value"
  ) +
  theme_minimal(base_size = 14) +
  theme(legend.position = "none")
```

- Hits tend to be more predictable in their energy and danceability
- Danceability does not clearly separate Hits from Non-Hits
- Energy seems more important → Hits are reliably energetic, while Non-Hits vary more widely

Regression Model

Call:
lm(formula = song_popularity ~ danceability * energy, data = song_data)

Residuals:
Min 1Q Median 3Q Max
-60.562 -12.484 2.716 15.846 46.214

Coefficients:
Estimate Std. Error t value Pr(>|t|)
(Intercept) 53.666 1.693 31.708 < 2e-16 ***
danceability -2.655 2.840 -0.935 0.35
energy -16.422 2.582 -6.359 2.08e-10 ***
danceability:energy 28.291 4.350 6.503 8.06e-11 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 21.76 on 18831 degrees of freedom
Multiple R-squared: 0.0131, Adjusted R-squared: 0.01295
F-statistic: 83.34 on 3 and 18831 DF, p-value: < 2.2e-16

term	estimate	conf.int	statistic	df	p.value
Intercept	53.67	[50.35, 56.98]	31.71	18831	< .001
Danceability	-2.65	[-8.22, 2.91]	-0.93	18831	.350
Energy	-16.42	[-21.48, -11.36]	-6.36	18831	< .001
Danceability $\times$ Energy	28.29	[19.76, 36.82]	6.50	18831	< .001

- Danceability alone does not predict popularity
- Energy alone predicts lower popularity
- High energy and high popularity in combination significantly increase popularity

```
# Load the song data dataset
song_data <- read.csv("song_data.csv")

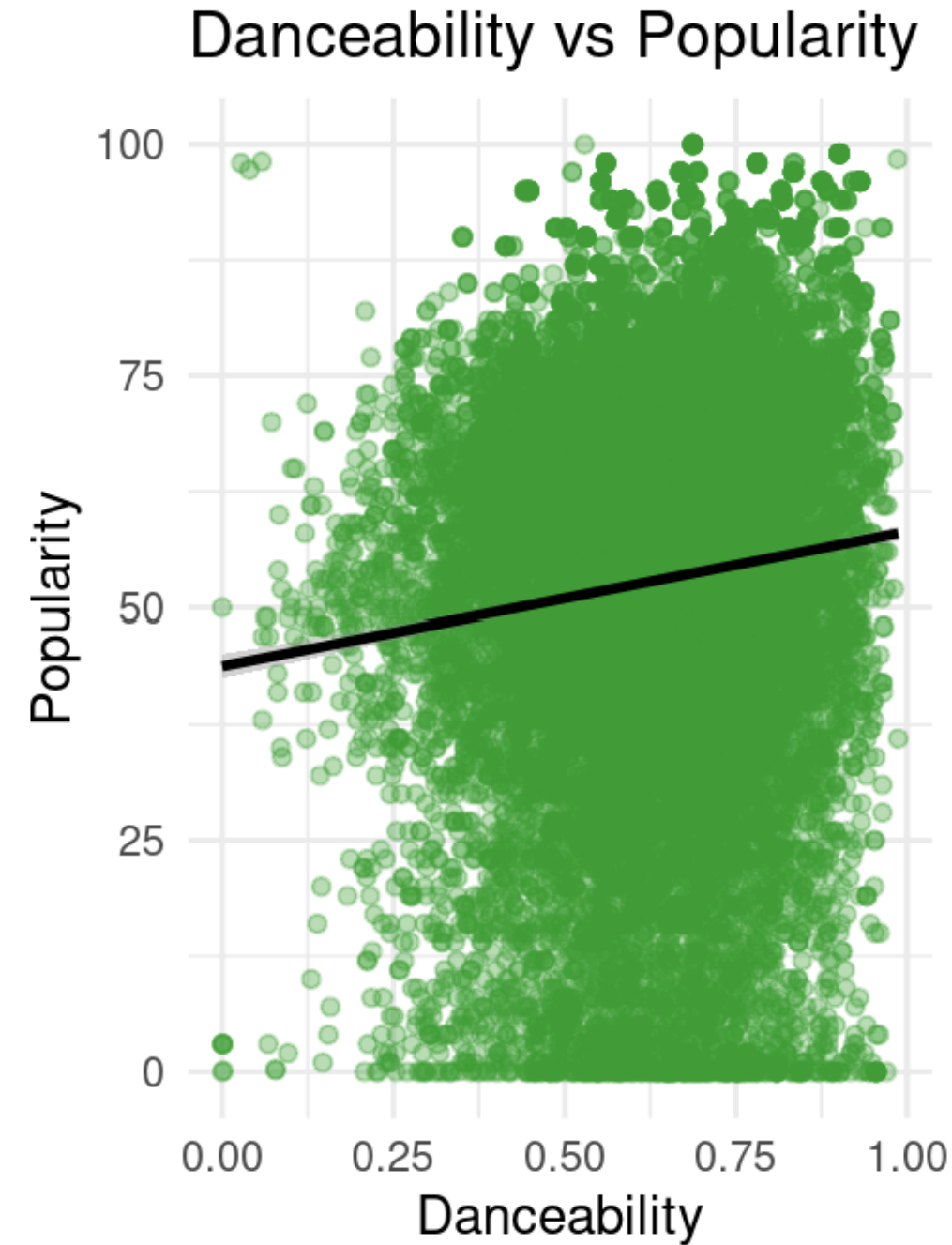
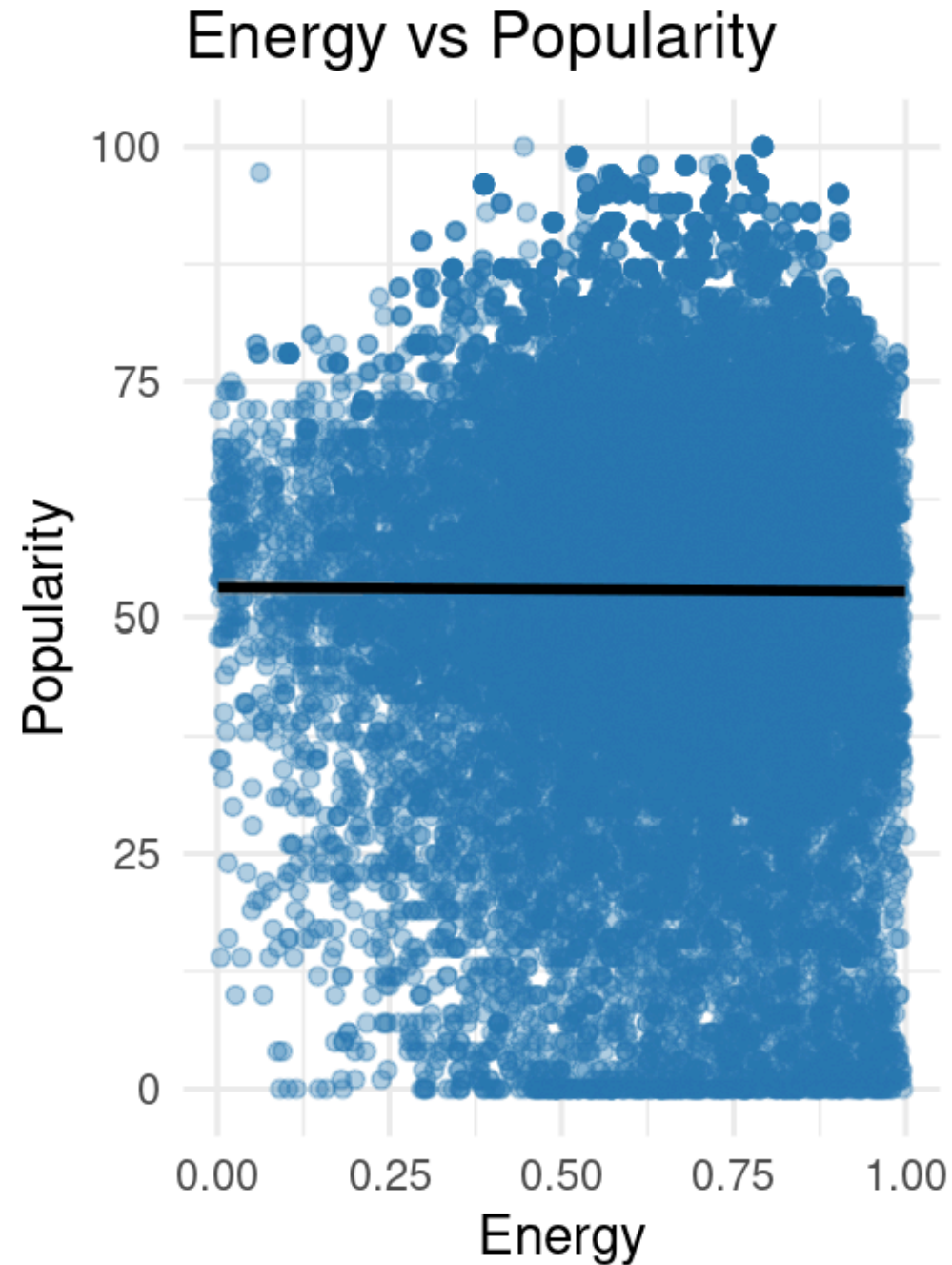
# overview
> str(song_data)

# Fit a linear regression model
m3 <- lm(song_popularity ~ danceability * energy, data = song_data)

# Summary of the model
summary(m3)

# create table
m3 <- lm(song_popularity ~ danceability * energy, data = song_data)
apa_lm <- apa_print(m3)
tt(apa_lm$table)
View(m3)
```

Popularity Trend Analysis (commercial success)



- Danceability increases more clearly across popularity levels
- Energy shows a weaker, more stable pattern
- Popularity is sufficiently varied for prediction analysis

Code:

```
library(tidyverse)
library(patchwork)

# Load data
spotify <- read_csv("song_data.csv")

# Clean and standardise values
spotify <- spotify %>%
  mutate(
    song_popularity = as.numeric(gsub("[^0-9.]", "", song_popularity)),
    danceability    = as.numeric(gsub("[^0-9.]", "", danceability)),
    danceability    = ifelse(danceability > 1, danceability / 100000, danceability),
    song_popularity = ifelse(song_popularity > 100,
                             song_popularity / 20,
                             song_popularity)
  )

# Remove invalid rows
spotify_clean <- spotify %>%
  filter(
    !is.na(song_popularity),
    !is.na(energy),
    !is.na(danceability),
    song_popularity >= 0 & song_popularity <= 100,
    danceability >= 0 & danceability <= 1
  )

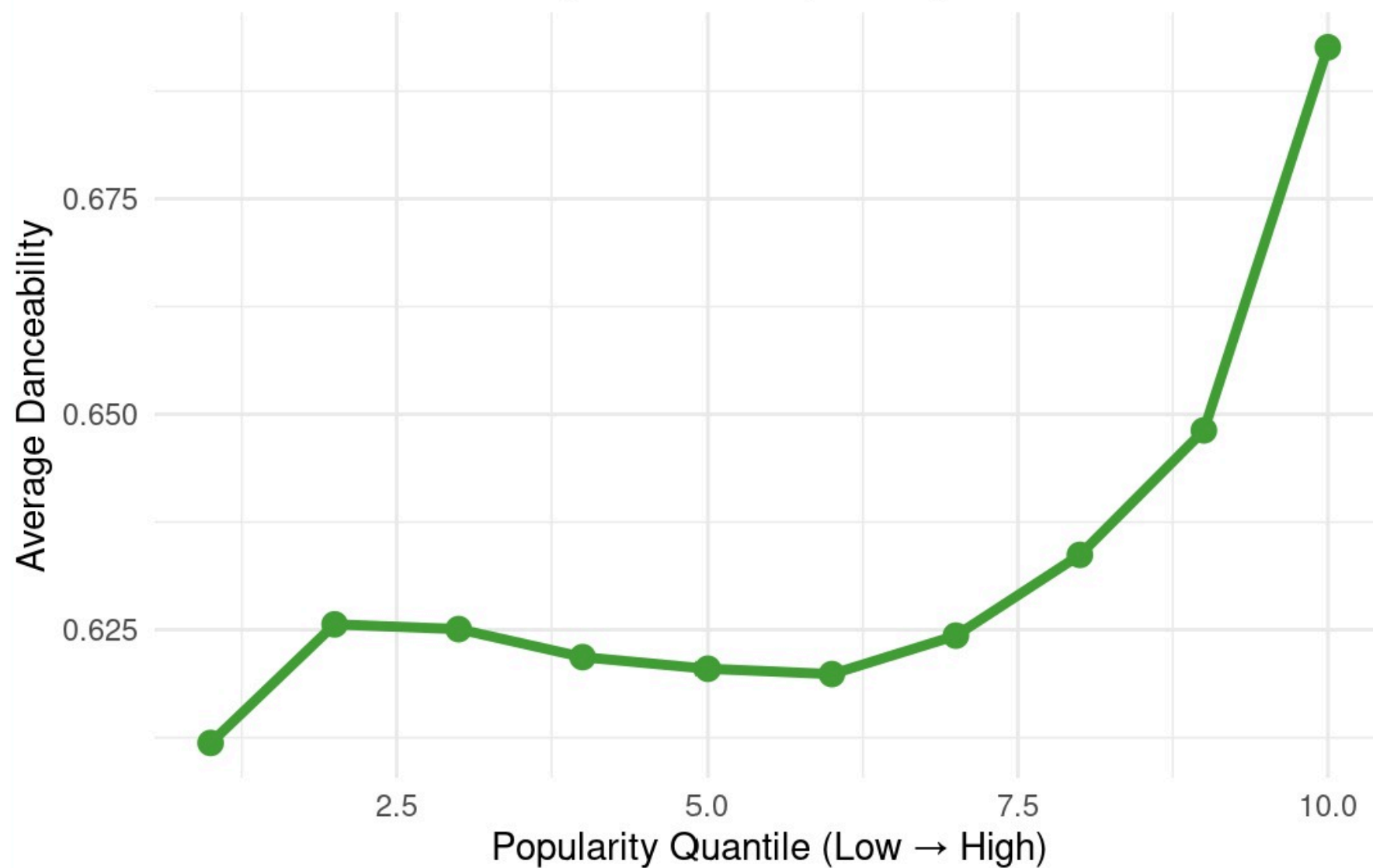
# Energy scatterplot
p_energy <- ggplot(spotify_clean, aes(energy, song_popularity)) +
  geom_point(color = "#1f78b4", alpha = 0.35, size = 1.4) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  theme_minimal() +
  labs(title = "Energy vs Popularity",
       x = "Energy", y = "Popularity")

# Danceability scatterplot
p_dance <- ggplot(spotify_clean, aes(danceability, song_popularity)) +
  geom_point(color = "#33a02c", alpha = 0.35, size = 1.4) +
  geom_smooth(method = "lm", color = "black", se = TRUE) +
  theme_minimal() +
  labs(title = "Danceability vs Popularity",
       x = "Danceability", y = "Popularity")

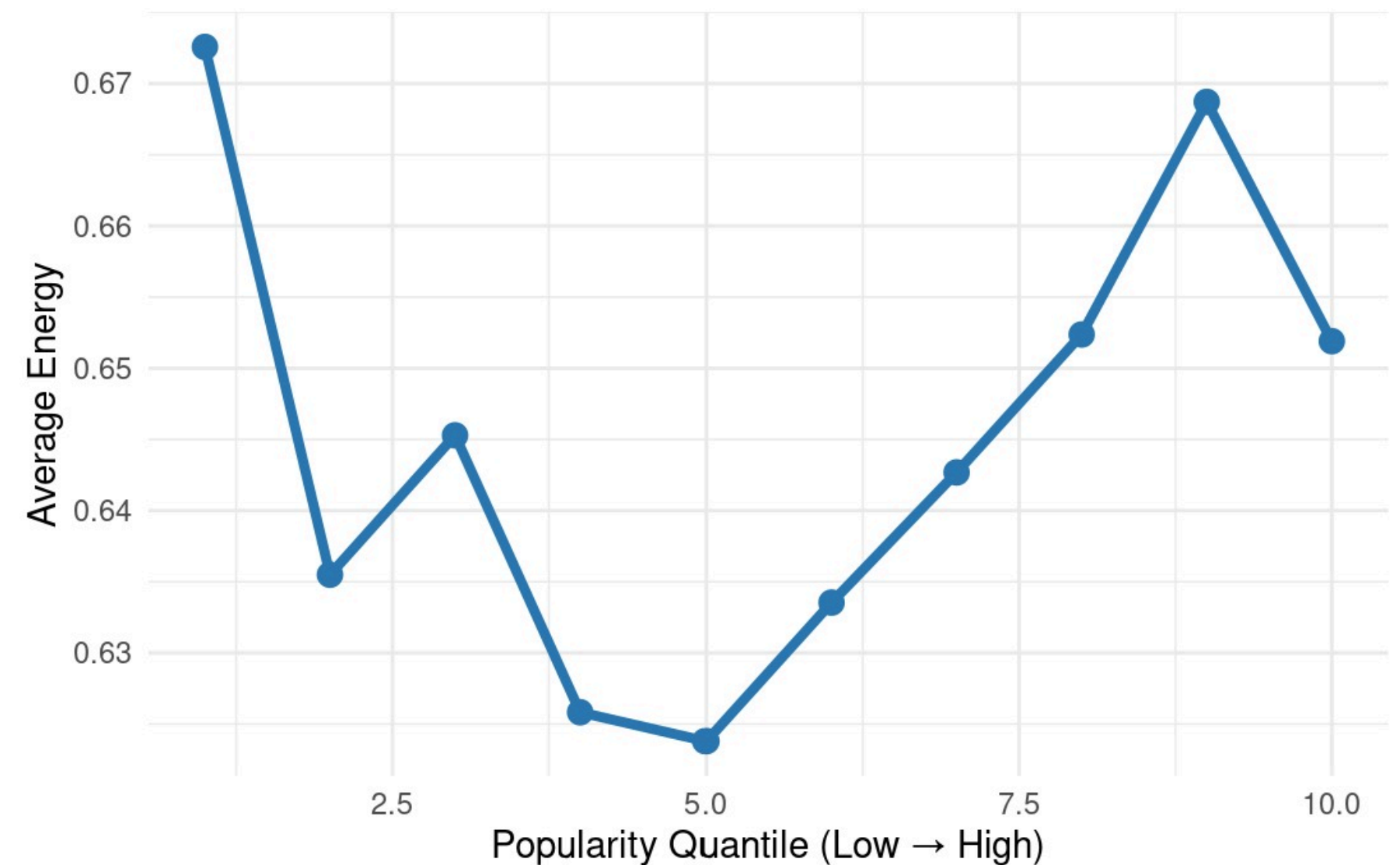
# Combine the two plots
p_energy + p_dance
```

How Audio Features Change With Increasing Popularity

Trend of Danceability Across Popularity Levels



Trend of Energy Across Popularity Levels



Code:

```
# Create popularity quantiles and group means
spotify_quantile <- spotify_clean %>%
  mutate(pop_quantile = ntile(song_popularity, 10)) %>%
  group_by(pop_quantile) %>%
  summarise(
    avg_dance = mean(danceability, na.rm = TRUE),
    avg_energy = mean(energy, na.rm = TRUE)
  )

# Danceability trend
ggplot(spotify_quantile, aes(pop_quantile, avg_dance)) +
  geom_line(color = "#33a02c", linewidth = 1.4) +
  geom_point(color = "#33a02c", size = 3) +
  theme_minimal() +
  labs(
    title = "Trend of Danceability Across Popularity Levels",
    x = "Popularity Quantile (Low → High)",
    y = "Average Danceability"
  )

# Energy trend
ggplot(spotify_quantile, aes(pop_quantile, avg_energy)) +
  geom_line(color = "#1f78b4", linewidth = 1.4) +
  geom_point(color = "#1f78b4", size = 3) +
  theme_minimal() +
  labs(
    title = "Trend of Energy Across Popularity Levels",
    x = "Popularity Quantile (Low → High)",
    y = "Average Energy"
  )
```

First conclusion

- **songs have enough variation in features for analysis**
- **Energy & danceability show positive connection with popularity**
- **Scatterplots confirm - trend exists, though not very strong**
- **Boxplots show: more popular songs - usually higher danceability & energy**
- **Regression: features matter; only explain small part → why a song becomes successful**
- **Trend analysis supports this: danceability increases as popularity increases, while energy stays stable**

Limitations

- Spotify API expires every 60 minutes → unstable for large data
- Required track-by-track calls → unrealistic for thousands of songs



API Constraints

Data Quality & Consistency Issues



- Genre strongly influences danceability + energy
- Release year affects popularity (e.g., old songs rank lower)



No Genre or Year Controls Yet

- The dataset may overrepresent popular, which could distort the relationship between audio features and commercial success.
- Missing or uneven genre distribution (e.g., Pop dominating the dataset) may bias the predictive patterns for danceability and energy



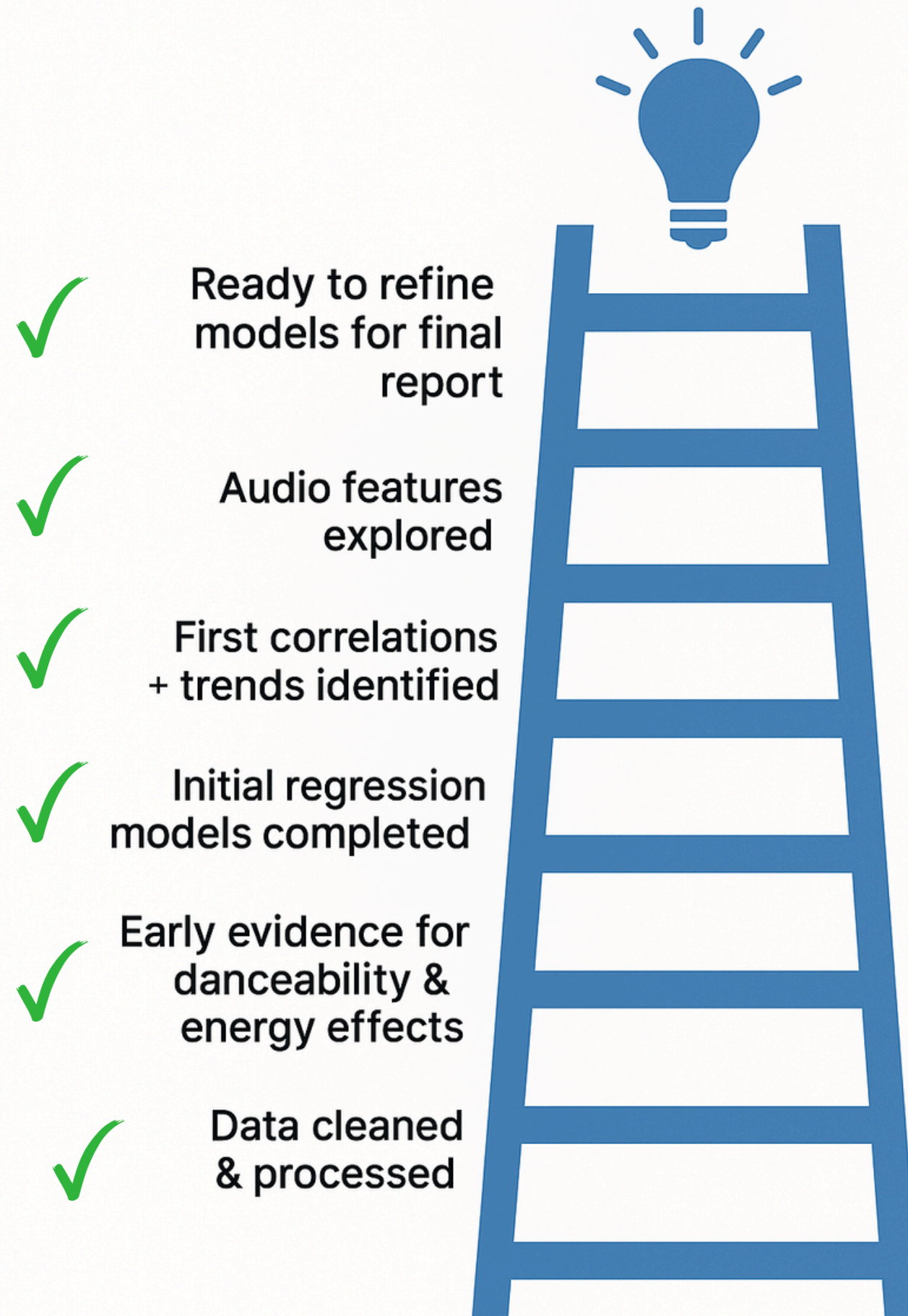
Potential Sampling Bias

Limited Predictive Power of Current Model

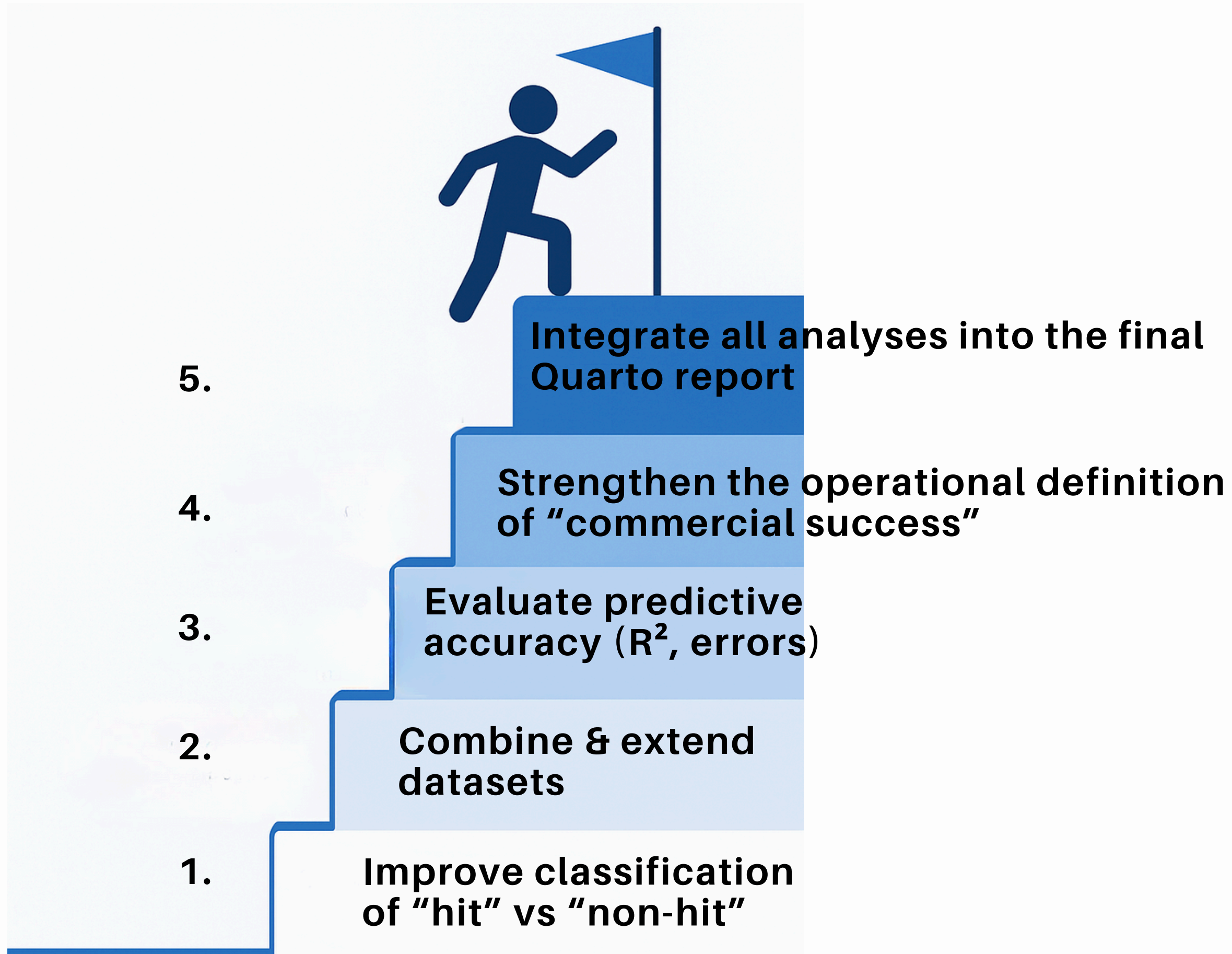


- Kaggle CSV exports contained inconsistent fields
- Missing values had to be cleaned manually
- R^2 is very low → audio features explain only a small part of popularity
- Success is influenced by external factors (marketing, playlists, virality)

Where are we now?



Next steps until final report



Thanks for your attention!!



Q & A

Sources/data utilized

- <https://www.kaggle.com/datasets/edalrami/19000-spotify-songs?resource=download>
- <https://tidyverse.org>
- <https://otexts.com>
- <https://www.kaggle.com/code/varunsaikanuri/spotif-data-visualization>